

DIGITAL SIGNAL PROCESSING – ECA0902

MS1 - JULY – 2026 – SLOT C – Keerthi Kassan V

Name: Saravana Yogesh M

Regno: 192412273

MATLAB EXPERIMENTS OUTPUT

EXPERIMENT 1

1.Butterworth IIR Filters

1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

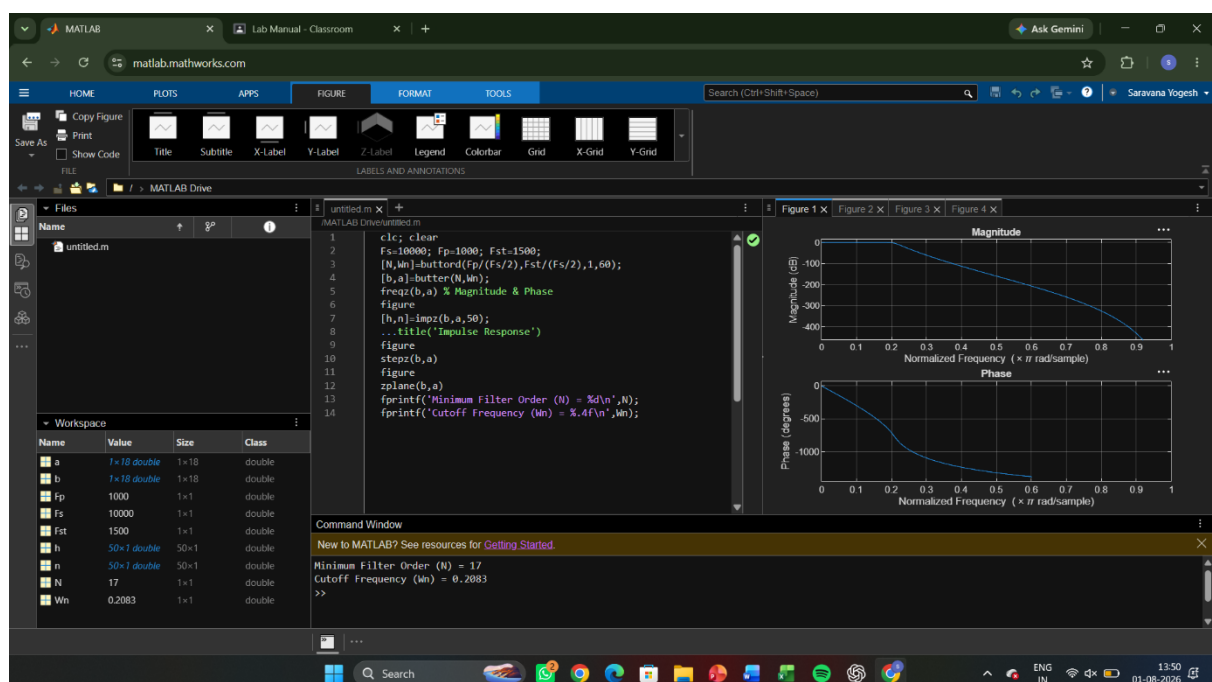
1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

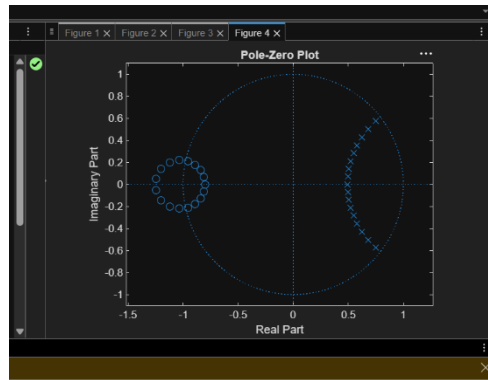
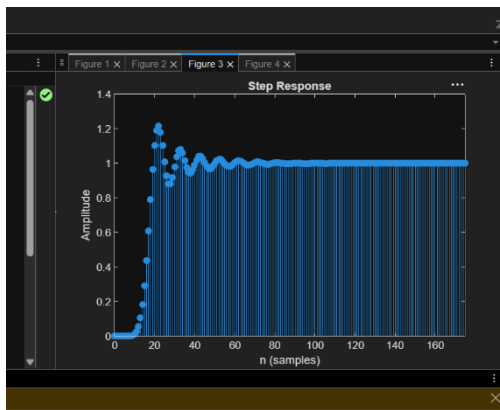
1.1)CODE:

```
clc; clear

Fs=10000; Fp=1000; Fst=1500;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn);
freqz(b,a) % Magnitude & Phase
figure
[h,n]=impz(b,a,50);
...title('Impulse Response')
figure
stepz(b,a)
figure
zplane(b,a)
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

OUTPUT:





1.2)CODE:

```
clc; clear

Fs=10000; Fp=3000; Fst=2000;

[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);

[b,a]=butter(N,Wn,'high');

% Magnitude & Phase

freqz(b,a)

% Impulse Response

figure

[h,n]=impz(b,a,50);

stem(n,h), grid on

title('Impulse Response')

% Step Response

figure

stepz(b,a)

title('Step Response')

% Group Delay

figure

grpdelay(b,a)

title('Group Delay')

% Pole-Zero Plot

figure
```

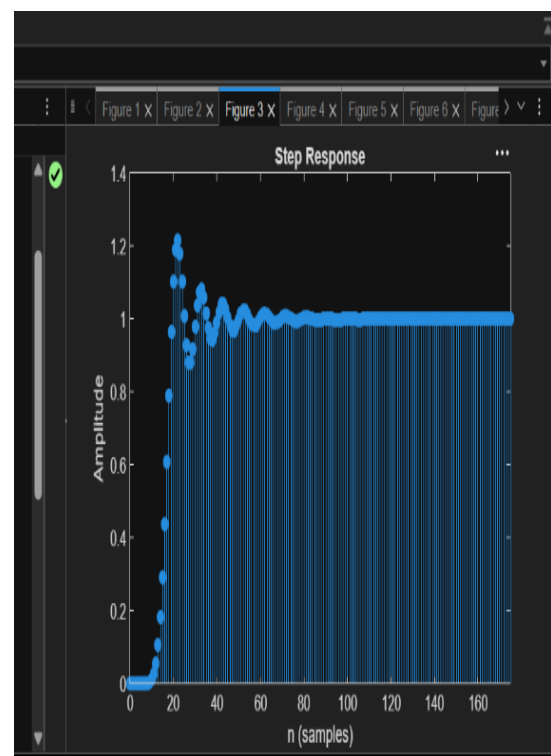
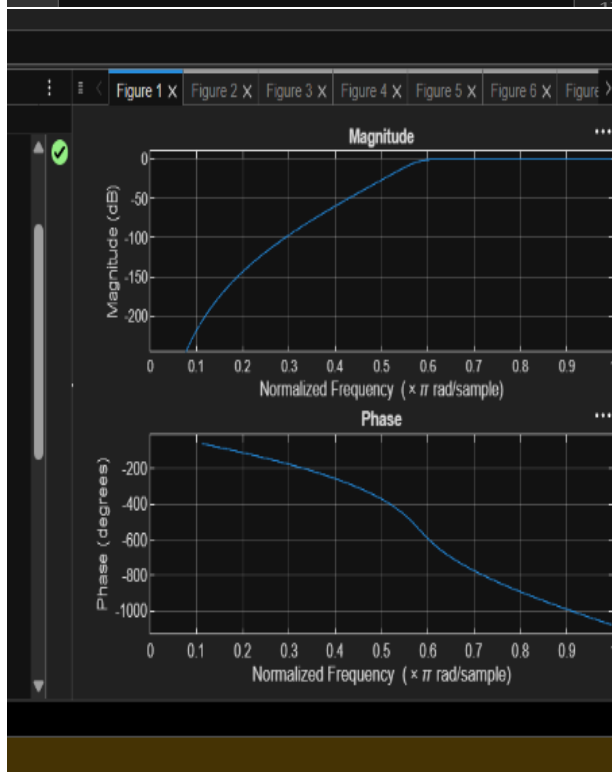
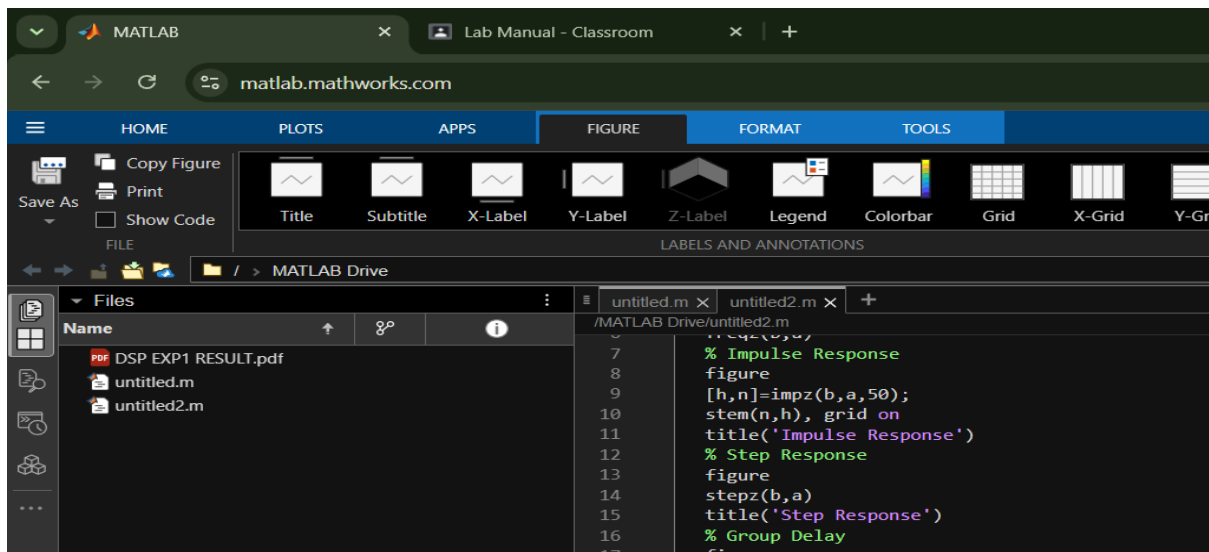
zplane(b,a)

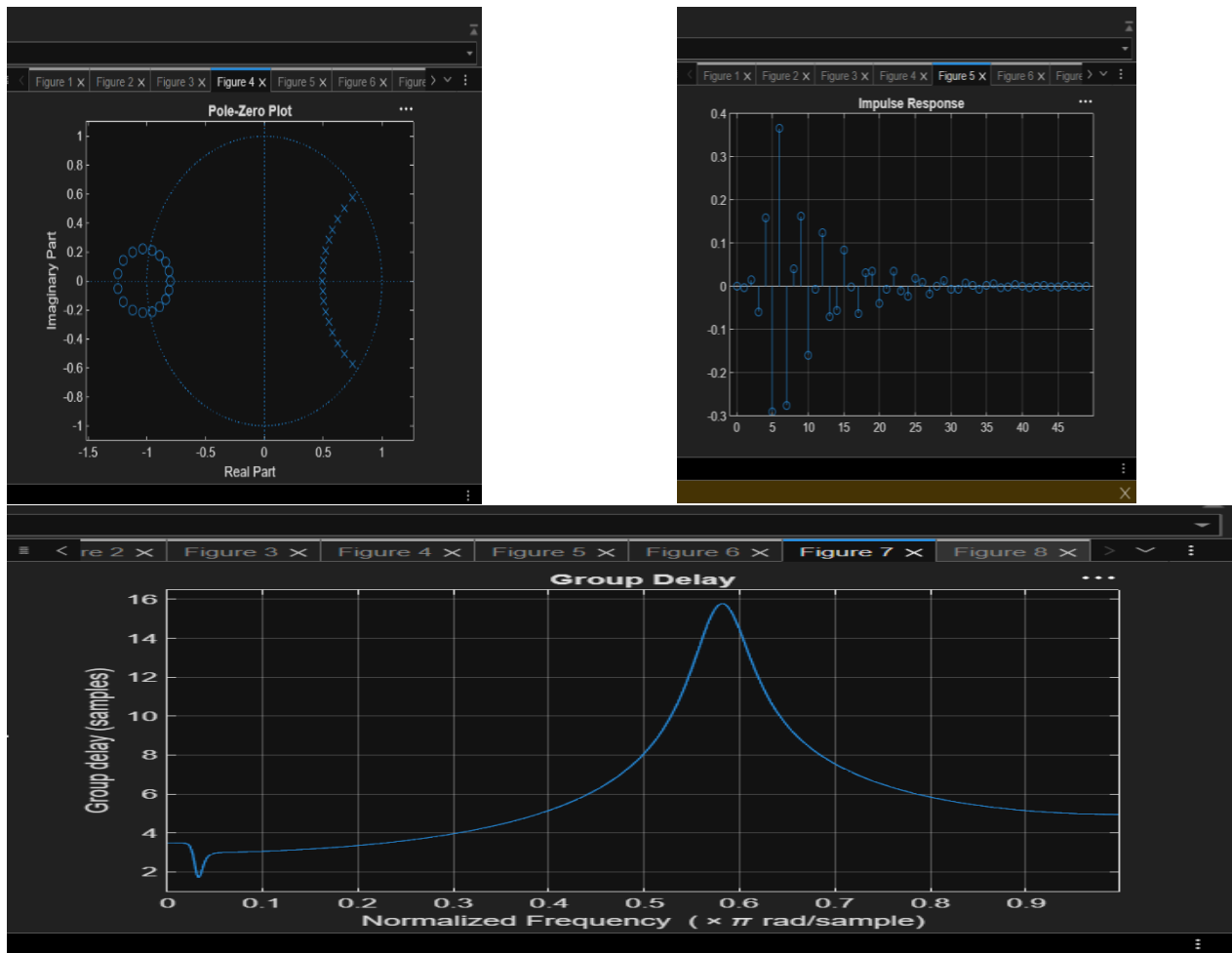
title('Pole-Zero Plot')

fprintf('Minimum Filter Order (N) = %d\n',N);

fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);

OUTPUT:





EXPERIMENT 2

2.Chebyshev IIR Filters

2.1 Design and performance Evaluation of a Chebyshev Type I IIR Low Pass Filter using MATLAB

2.2 Design and performance Evaluation of a Chebyshev Type I IR High Pass Filter using MATLAB

2.1) CODE:

```
clc; clear;

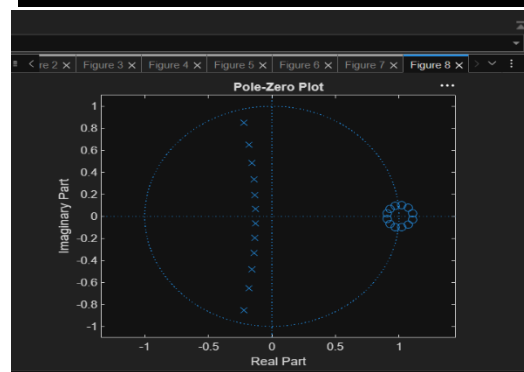
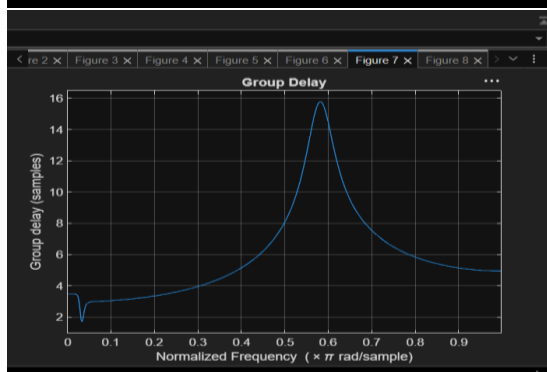
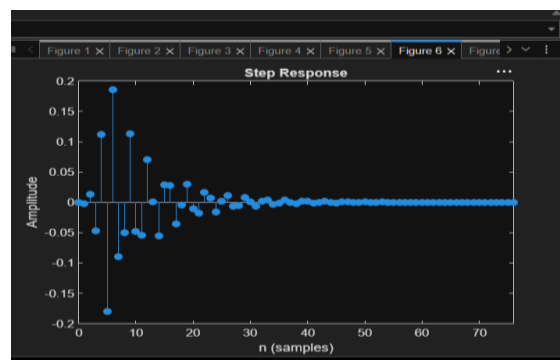
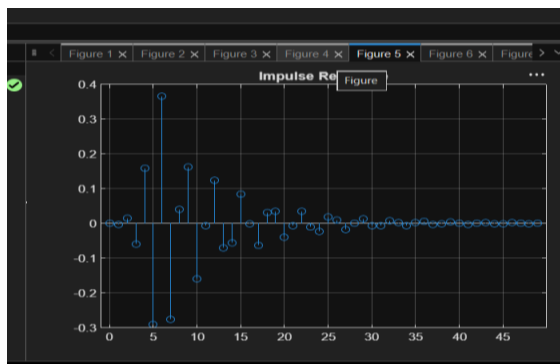
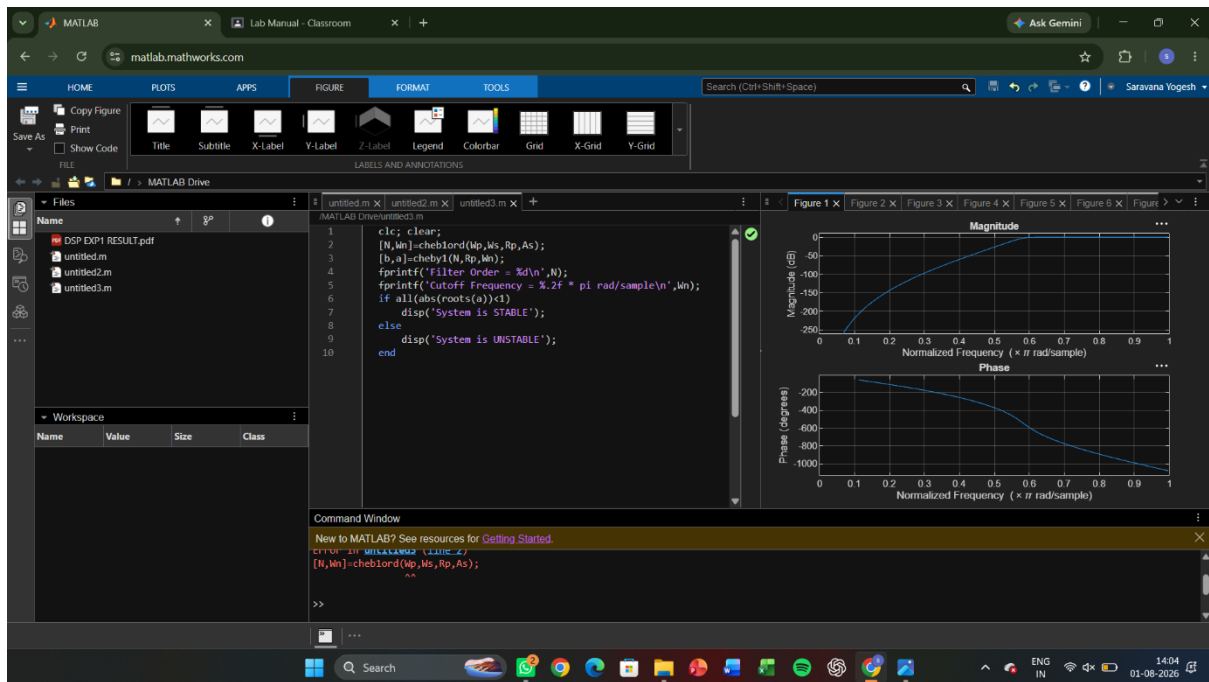
[N,Wn]=cheb1ord(Wp,Ws,Rp,As);
[b,a]=cheby1(N,Rp,Wn);
fprintf('Filter Order = %d\n',N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);
if all(abs(roots(a))<1)
```

```

disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

```

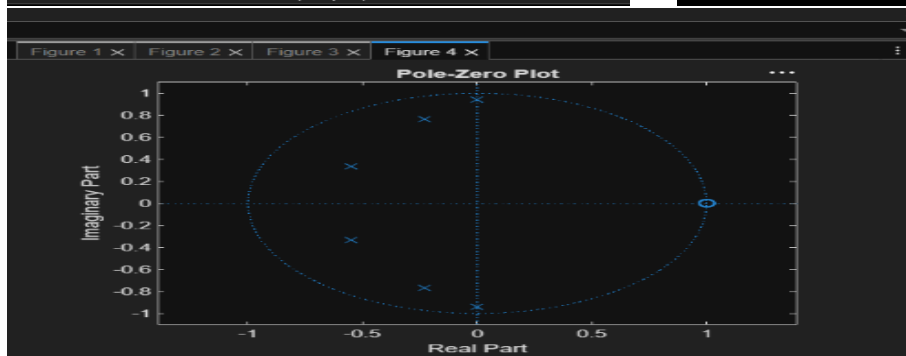
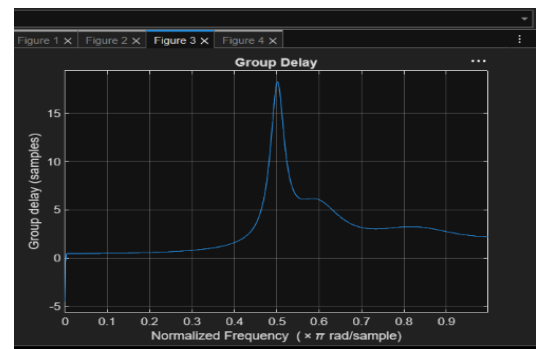
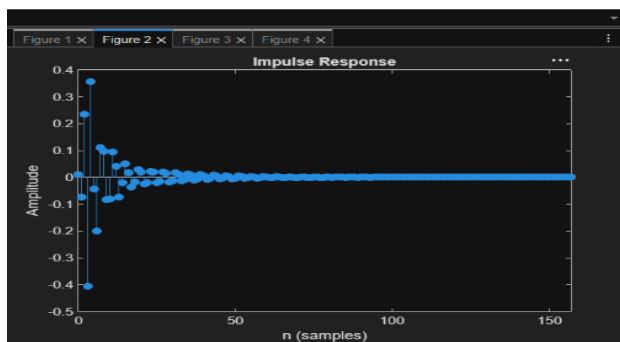
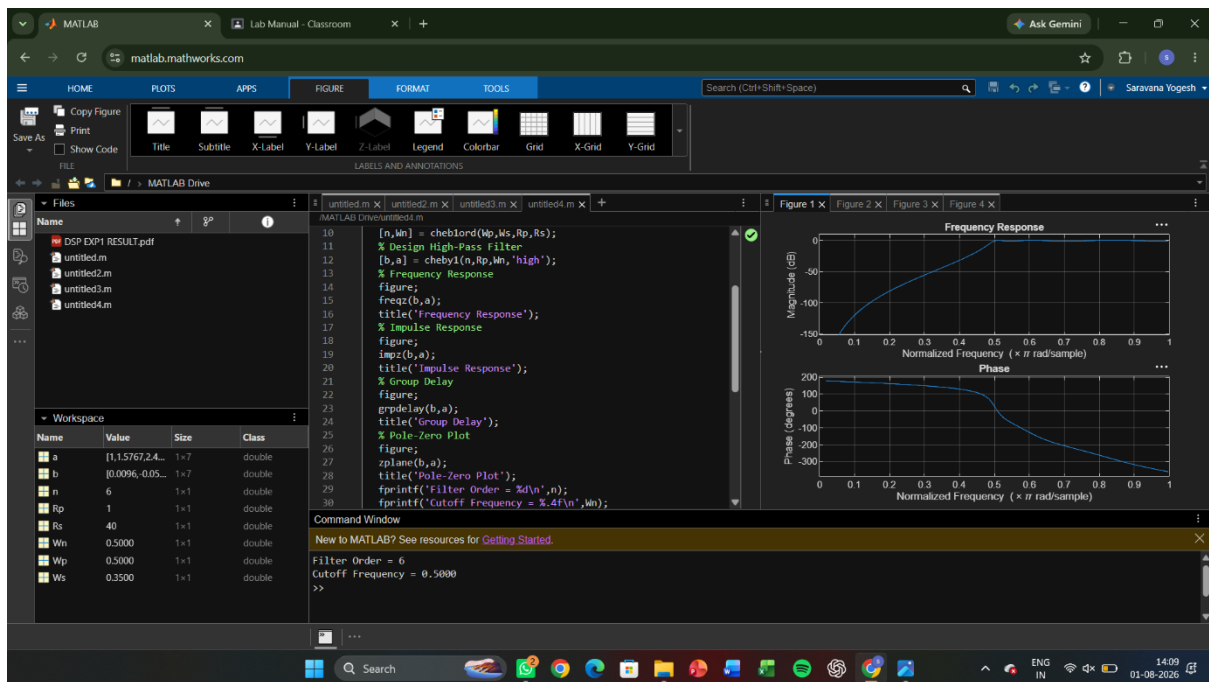
OUTPUT



2.2)CODE:

```
clc;
clear;
close all;
% Specifications
Rp = 1; % Passband Ripple (dB)
Rs = 40; % Stopband Attenuation (dB)
Wp = 0.5; % Passband Edge
Ws = 0.35; % Stopband Edge
% Find Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);
% Design High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');
% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');
% Impulse Response
figure;
impz(b,a);
title('Impulse Response');
% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');
% Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');
fprintf('Filter Order = %d\n',n);
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

OUTPUT:



EXPERIMENT 3

3. Performance Analysis of IIR Filters.

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

3.1) CODE

```
clc; clear; close all;

% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);

% Butterworth Filter
[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');

% Chebyshev Type-I Filter
[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');

% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');
figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

% Group Delay
figure;
grpdelay(b1,a1); hold on;
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');

% Pole-Zero Plot
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')

% Impulse Response
figure;
subplot(211),impz(b1,a1),title('Butterworth Impulse')
subplot(212),impz(b2,a2),title('Chebyshev Impulse')

% Pole Magnitude
figure;
```

```
subplot(121),stem(abs(roots(a1))),'filled')

title('Butterworth Pole Magnitude')

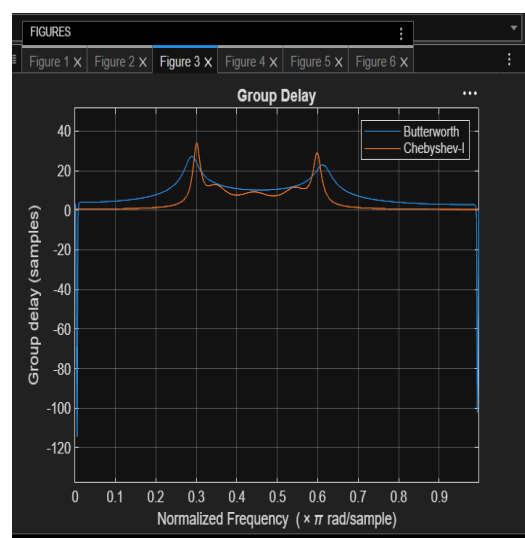
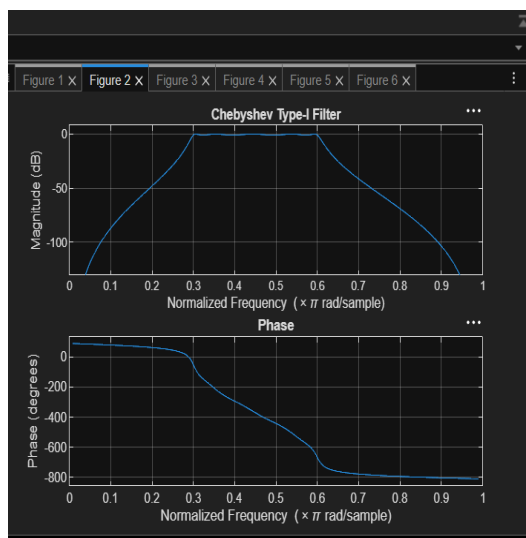
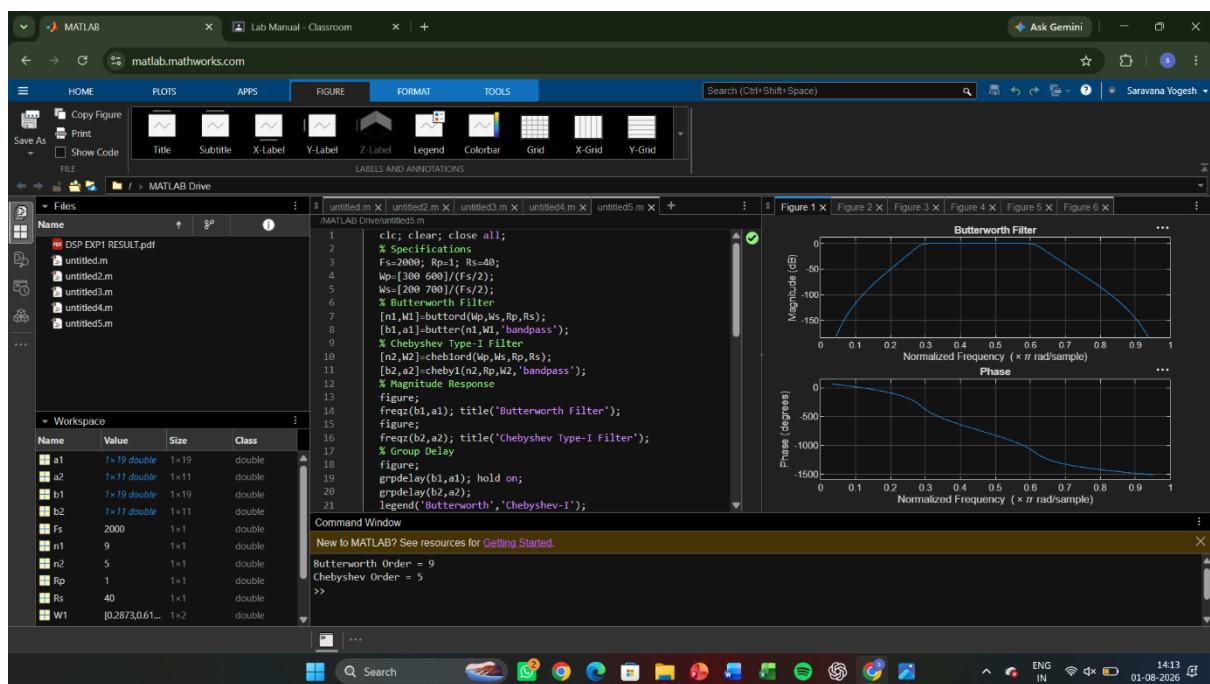
subplot(122),stem(abs(roots(a2))),'filled')

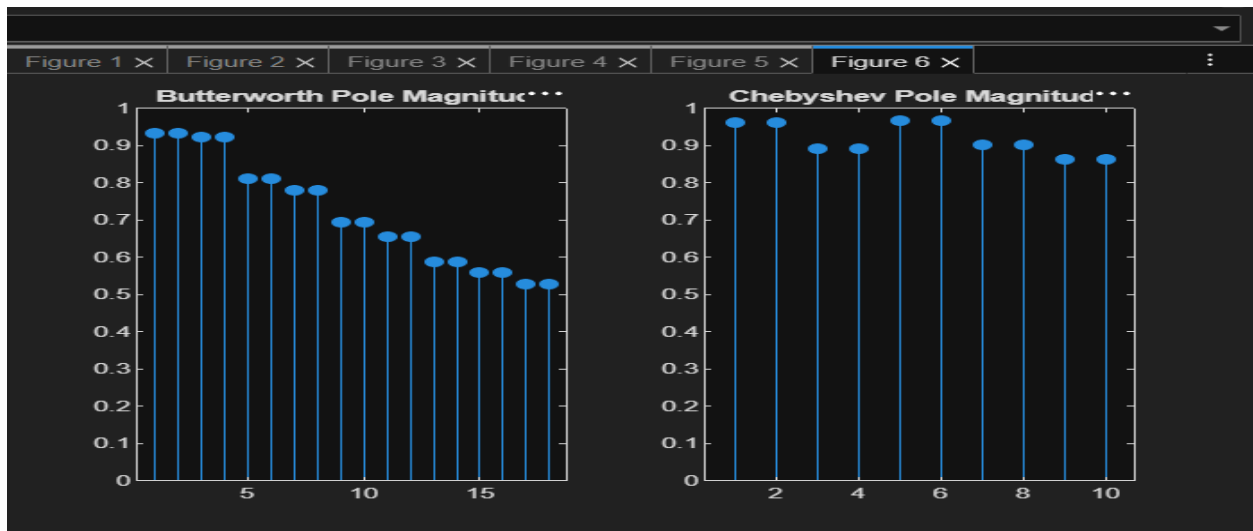
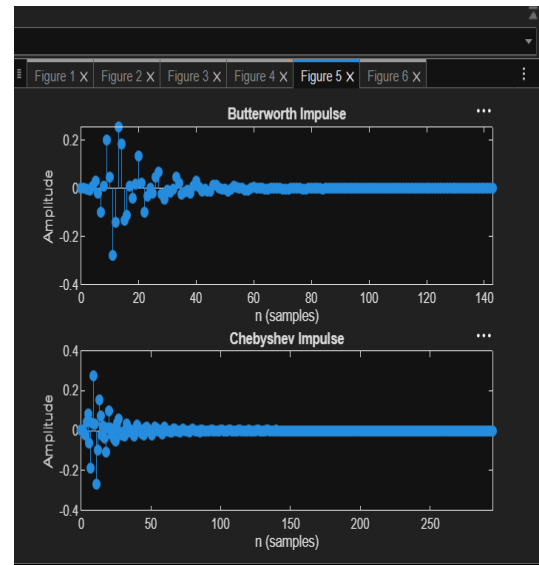
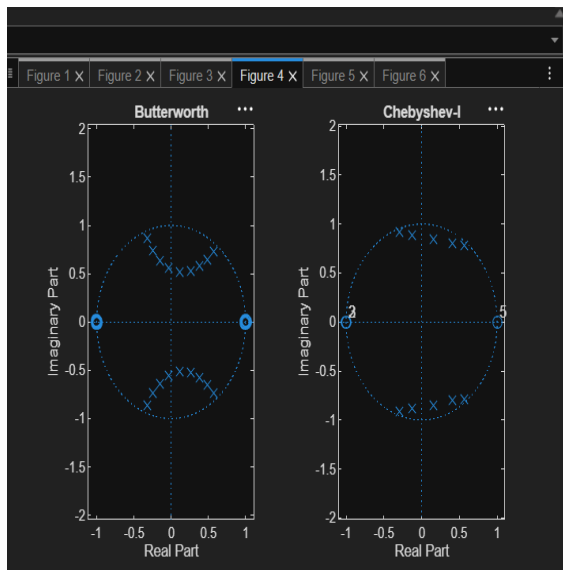
title('Chebyshev Pole Magnitude')

disp(['Butterworth Order = ',num2str(n1)])

disp(['Chebyshev Order = ',num2str(n2)])
```

OUTPUT:





EXPERIMENT 4

4. FIR Low Pass FilterS using Windowing Techniques

4.1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB.

4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique in MATLAB.

4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB.

4.1)CODE:

```
clc;

clear;

close all;

%%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%%% --- 1. FIR Low Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'low', rectwin(N+1));

disp('--- Rectangular Window Coefficients ---');

disp(b_rect);

%%% --- 2. FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm);

%%% --- 3. FIR Low Pass using Hanning Window ---

b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');

disp(b_hann);

%%% --- Plot Frequency Responses ---

figure;

freqz(b_rect,1);

title('Low-pass FIR using Rectangular Window');

figure;

freqz(b_hamm,1);

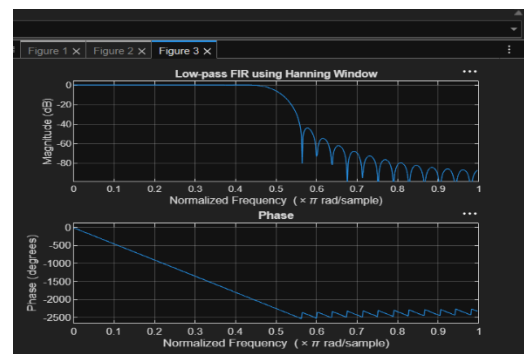
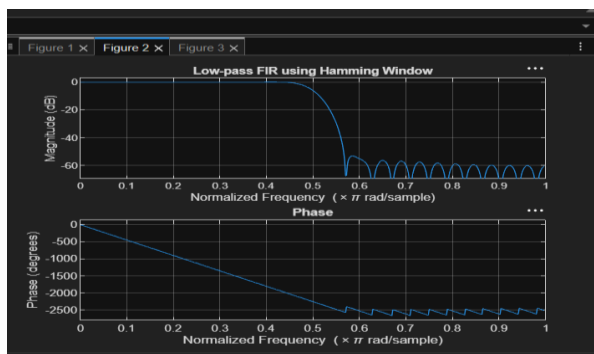
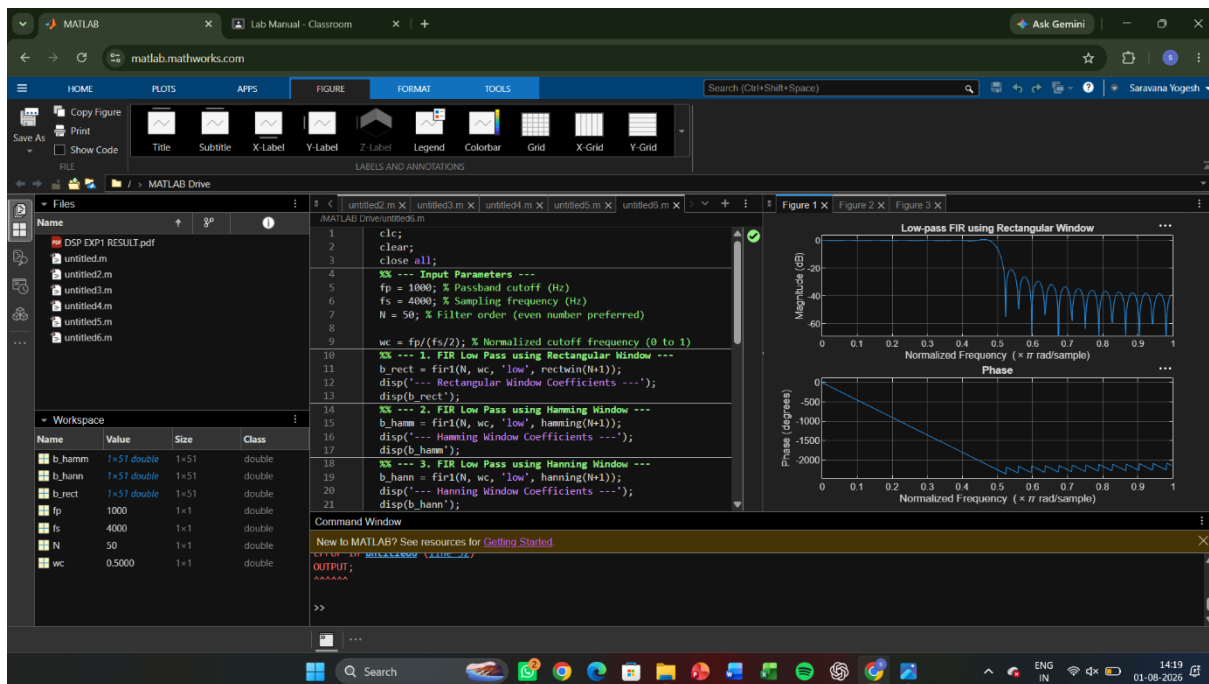
title('Low-pass FIR using Hamming Window');

figure;

freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');
```

OUTPUT;



4.2) CODE:

```
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hamming Window ---
```

```

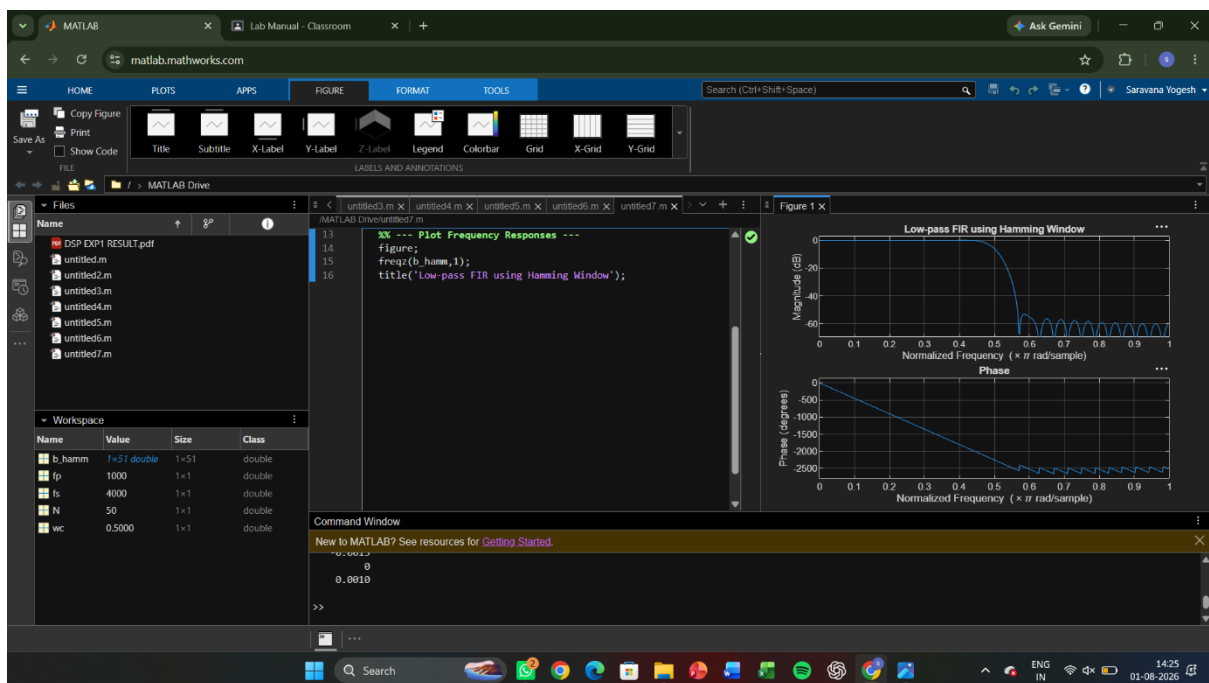
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm);

%% --- Plot Frequency Responses ---

figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');

```

OUTPUT



4.3)CODE:

```

clc;
clear;
close all;

%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

```

```
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
```

```
%% --- FIR Low Pass using Hanning Window ---
```

```
b_hann = fir1(N, wc, 'low', hanning(N+1));
```

```
disp('--- Hanning Window Coefficients ---');
```

```
disp(b_hann');
```

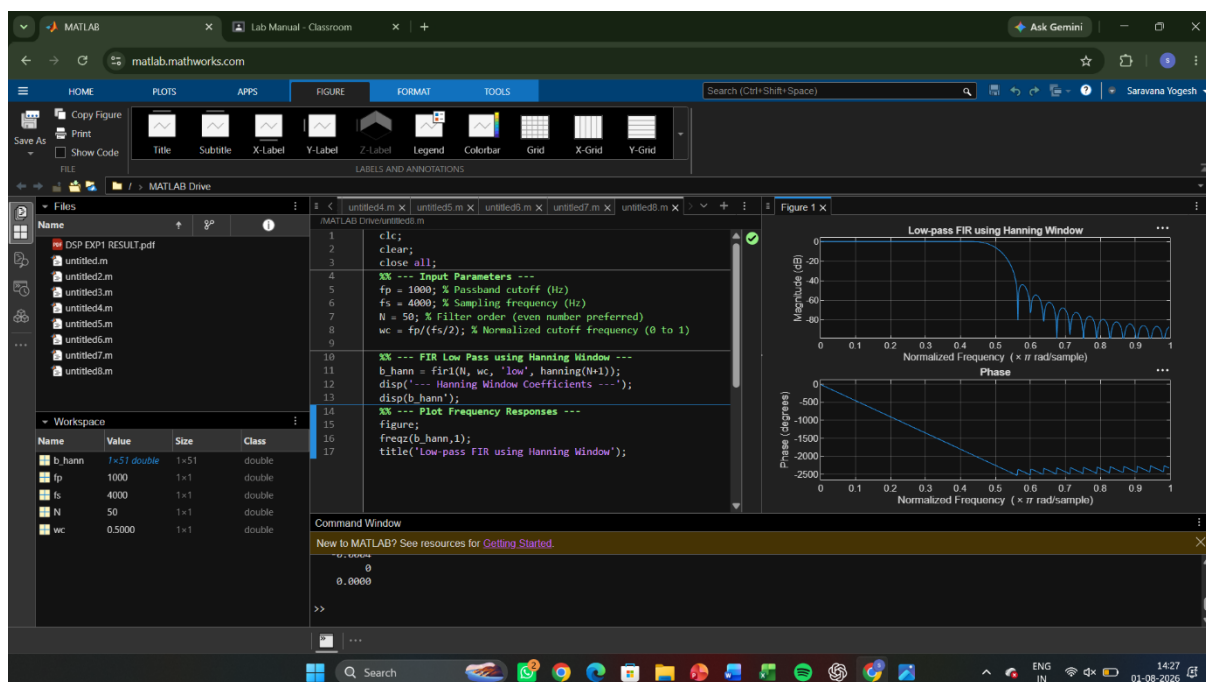
```
%% --- Plot Frequency Responses ---
```

```
figure;
```

```
freqz(b_hann,1);
```

```
title('Low-pass FIR using Hanning Window');
```

OUTPUT



EXPERIMENT 5

5. FIR High Pass FilterS using Windowing Techniques

5.1. Design and Analysis of an FIR High-Pass Filter Using the Rectangular Window Technique in MATLAB.

5.2. Design and Analysis of an FIR High-Pass Filter Using the Hamming Window Technique in MATLAB. .

5.3. Design and Analysis of an FIR High-Pass Filter Using the Hanning Window Technique in MATLAB.

5.1)CODE

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- 1. FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);

%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

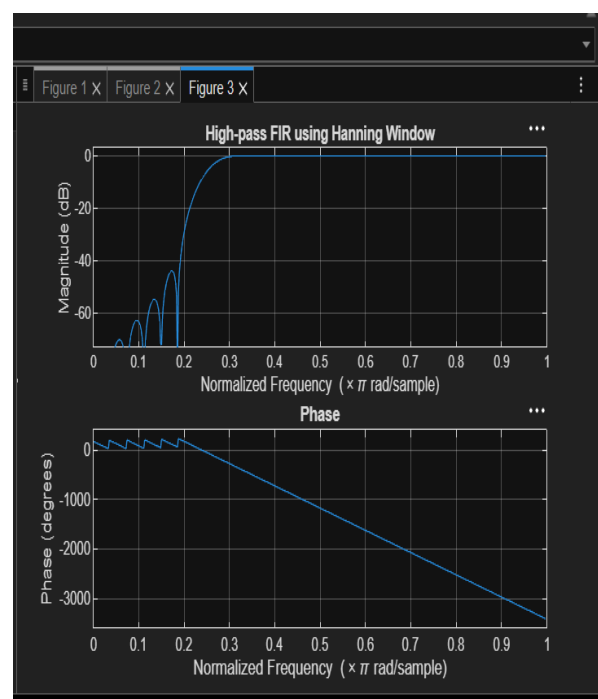
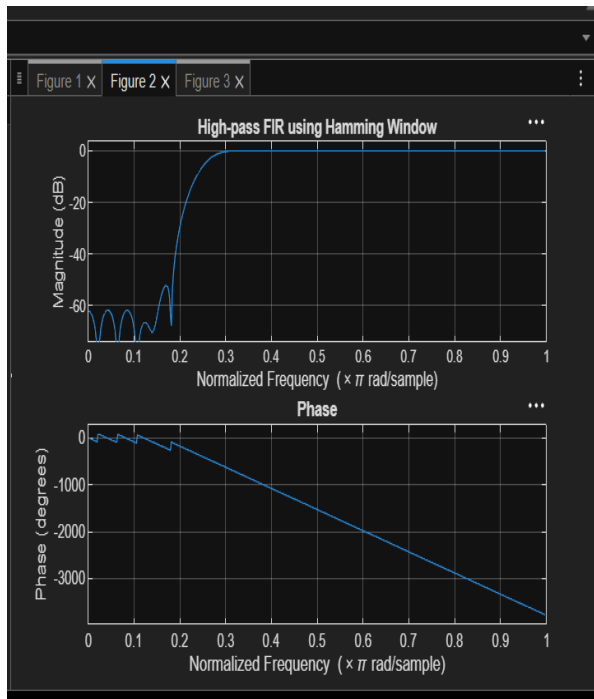
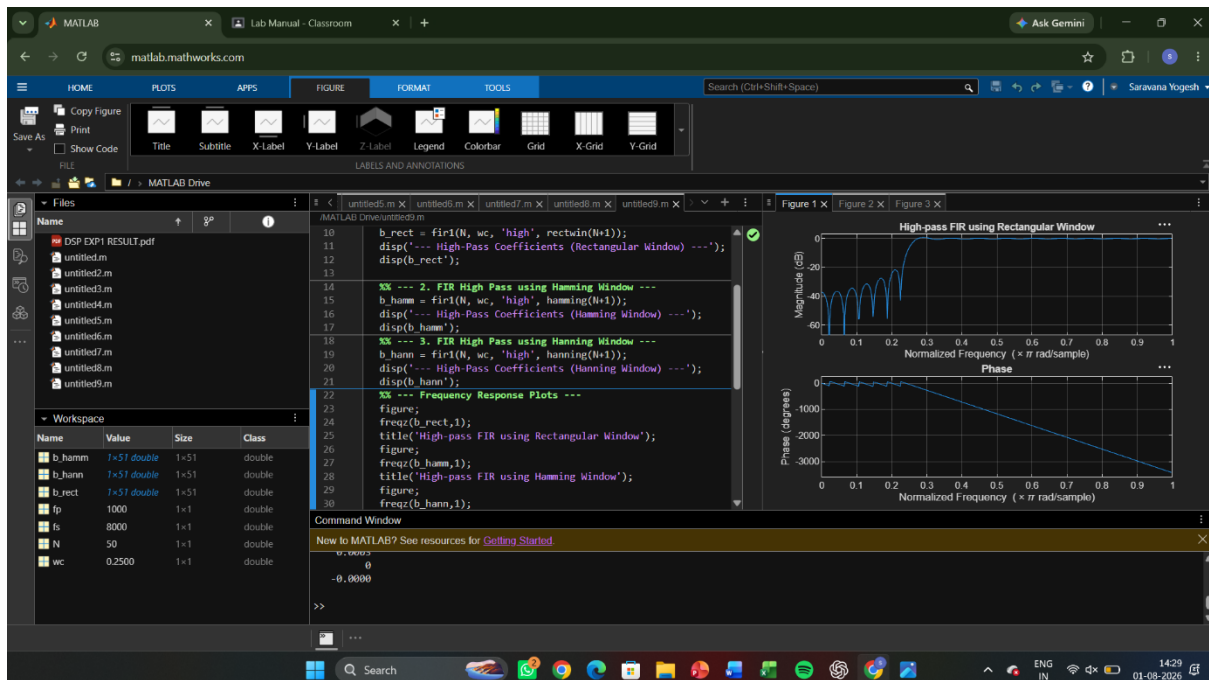
%% --- Frequency Response Plots ---
figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
```

figure;

freqz(b_hann,1);

title('High-pass FIR using Hanning Window');

OUTPUT



5.2)CODE

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)

fs = 8000; % Sampling frequency (Hz)

N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');

disp(b_hamm);

%% --- Frequency Response Plots ---

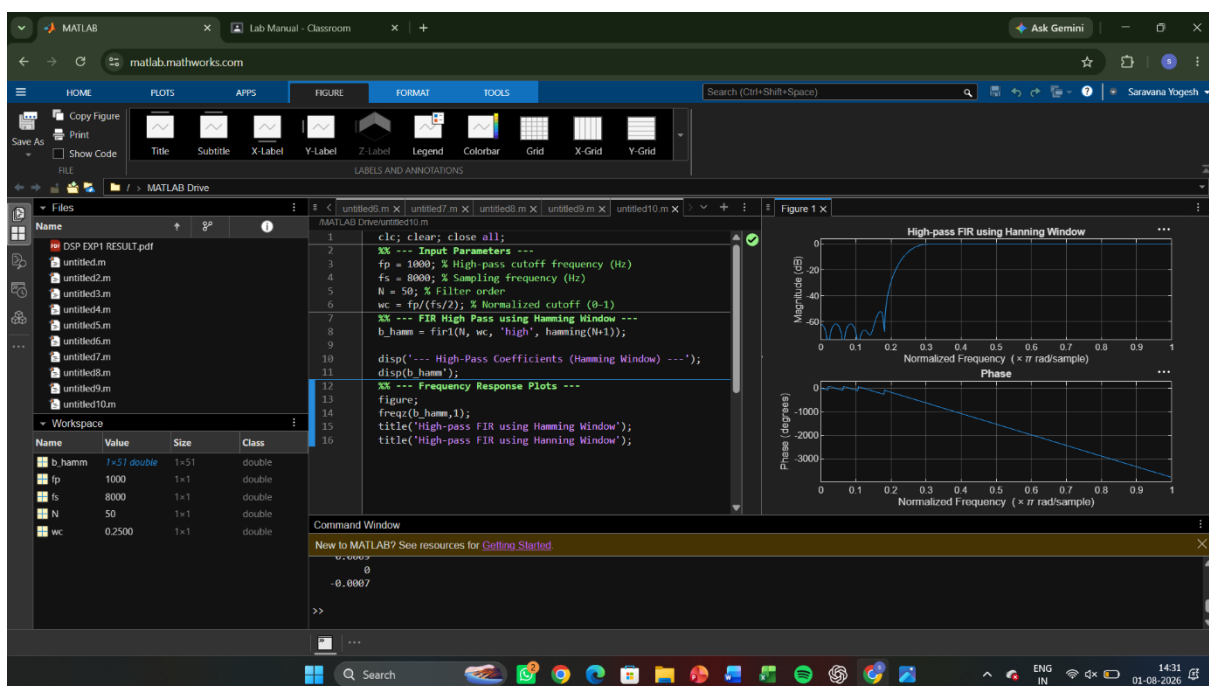
figure;

freqz(b_hamm,1);

title('High-pass FIR using Hamming Window');

title('High-pass FIR using Hanning Window');
```

OUTPUT:



5.3) CODE

```
clc;

clear;

close all;

%% Signal
fs = 1000; f = 50;

t = 0:1/fs:0.1;

x = sin(2*pi*f*t);

%% Quantization Levels
L = [8 16 32];

colors = {'r','g','m'};

figure;

for i = 1:length(L)

    % Plot Quantized Signals

    subplot(4,1,i+1);

    stairs(t,xq{i},colors{i},'LineWidth',1.2);

    title(['Quantized Signal ( ' num2str(L(i)) ' Levels)']);

    grid on;

end

% Original Signal

subplot(4,1,1);

plot(t,x,'LineWidth',1.5);

title('Original Signal'); grid on;

% Display MSE

fprintf('MSE for 8 Levels = %f\n', mse(1));

fprintf('MSE for 16 Levels = %f\n', mse(2));

fprintf('MSE for 32 Levels = %f\n', mse(3));

%% Error Plots

figure;

for i = 1:3

    subplot(3,1,i);
```

```

plot(t,e{i});

title(['Error (' num2str(L(i)) ' Levels)']);

grid on;

end

```

OUTPUT:

